Stability of MIMO Feedback Loops: Characteristic Loci

Multivariable Control, Homework 7

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English summary of my original Persian report, which is kept as a PDF in this folder.

Task

Problems 5-1 to 5-5 of Khaki-Sedigh, *Multivariable Control* (slide 66, stability).

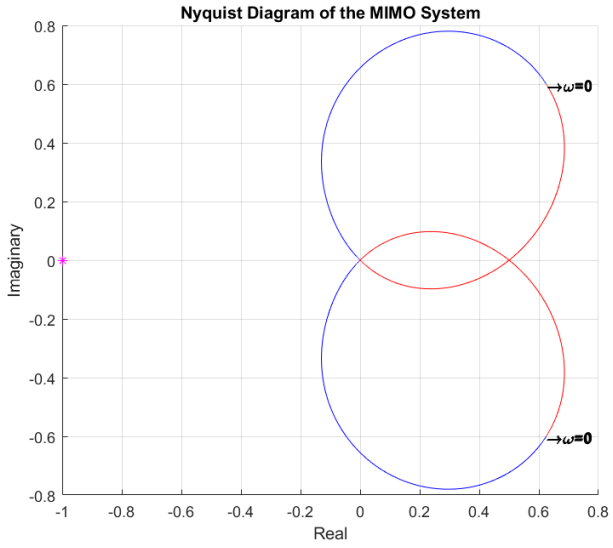
What I did

For the plant

$$G(s) = \begin{bmatrix} \dfrac{1}{s+1} & \dfrac{-1}{s+2} \\ \dfrac{1}{s+1} & \dfrac{1}{s+4} \end{bmatrix},$$

I wrote a MATLAB script that computes the eigenvalues of $G(s)$ symbolically and evaluates them along the imaginary axis to draw the characteristic loci, the generalised Nyquist plot. I then applied the generalised Nyquist criterion. I repeated the plots for loops with the controller gains K_1 and K_2 to see how the loci move relative to the critical point -1.

The script was only printed in the report. For this repository I copied it into `char_loci.m` unchanged and re-ran it in R2023b:



Neither locus encircles -1, and the plant has no right-half-plane poles, so by the generalised Nyquist criterion the unity-feedback loop is stable.